



Developing, assessing and visualizing FAIR metrics for biomedical datasets in HuBMAP



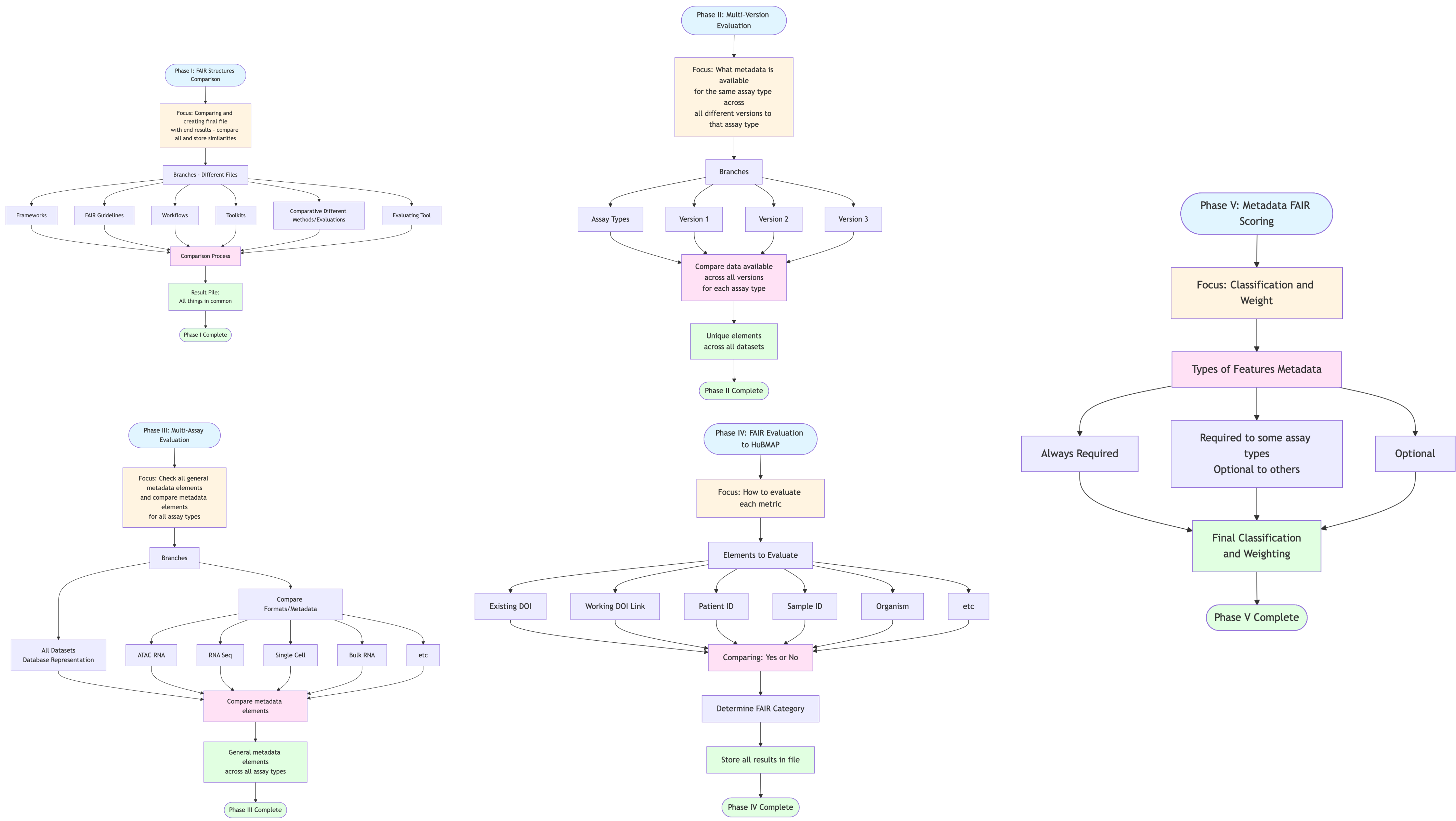
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ABSTRACT

As science evolves, researchers face growing challenges in handling large-scale datasets, complex workflows, and effective visualization. The Human BioMolecular Atlas Program (HuBMAP) was developed to address these challenges by providing a central platform to store, analyze, and share data. However, making data available is not enough—ensuring that it is Findable, Accessible, Interoperable, and Reusable (FAIR) is critical for maximizing its value to the research community. The importance of FAIRness lies in transforming datasets from static collections into resources that can be reliably discovered, understood, and reused. In HuBMAP, FAIRness is not only a claim but a measurable quality. A key difficulty was the absence of a static list of FAIR metrics; existing resources offered only broad guidelines, and every project has unique needs. To address this, we reviewed published FAIRness frameworks, evaluated their criteria, and synthesized a set of tailored FAIR metrics for HuBMAP. These metrics were organized into the four FAIR categories—Findable, Accessible, Interoperable, and Reusable—and applied to the different dataset versions and assay types within the project. This framework enables assessment from multiple perspectives: Users can evaluate dataset quality without exposing sensitive information, Uploaders gain guidance on how to improve completeness and transparency for each dataset, Curators can verify gaps and track progress and Administrators can monitor overall FAIRness across the system. To complement the metrics, HuBMAP also incorporates graphical tools that display how datasets score within each FAIR category. These visualizations provide a clear overview of dataset performance and allow exploration by HuBMAP ID, by institution, or across the entire collection. In summary, FAIRness metrics in HuBMAP were designed and implemented to ensure data is not only available but also usable, transparent, and valuable for advancing biomedical research.

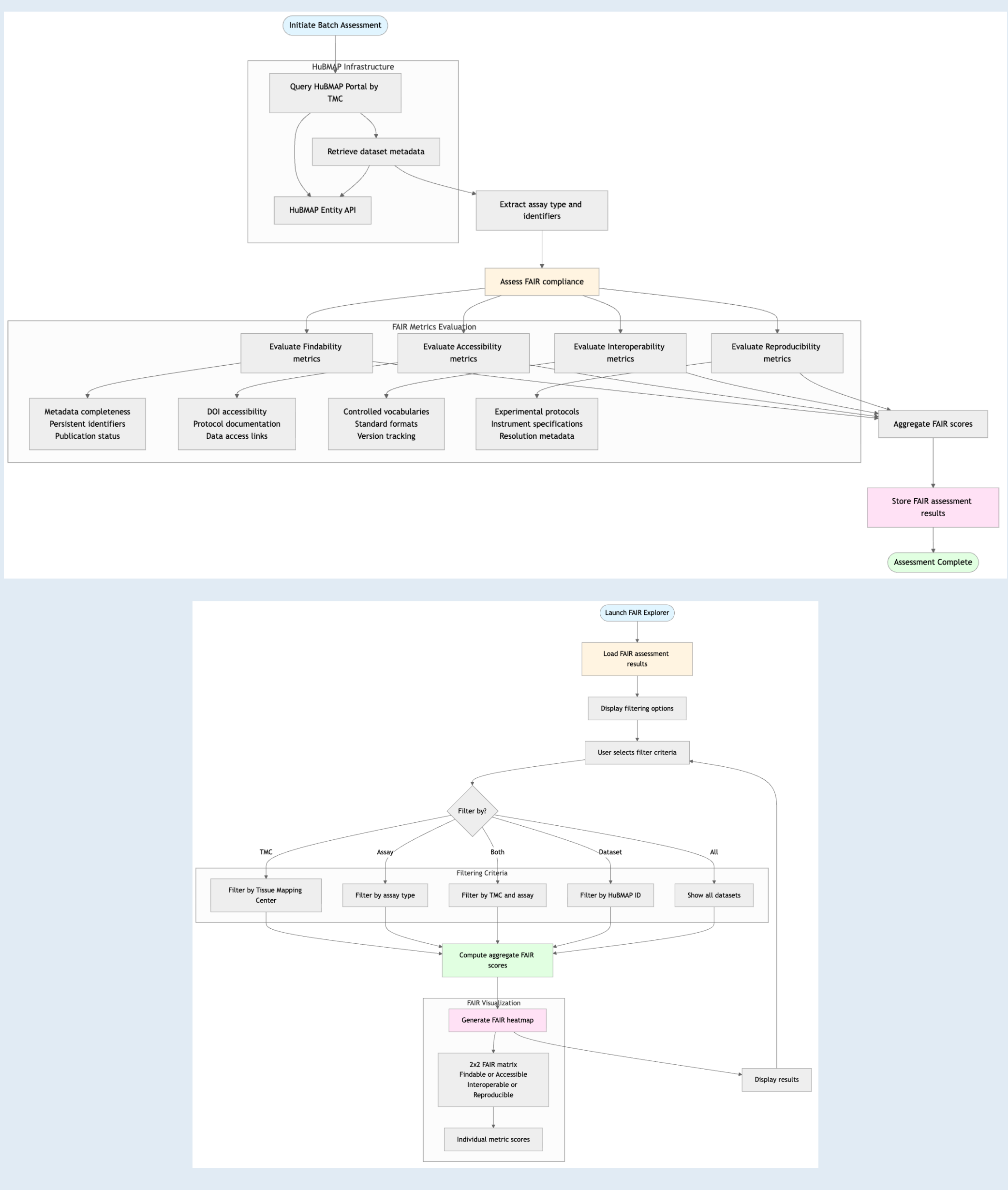
INTRODUCTION

Infrastructure Overview

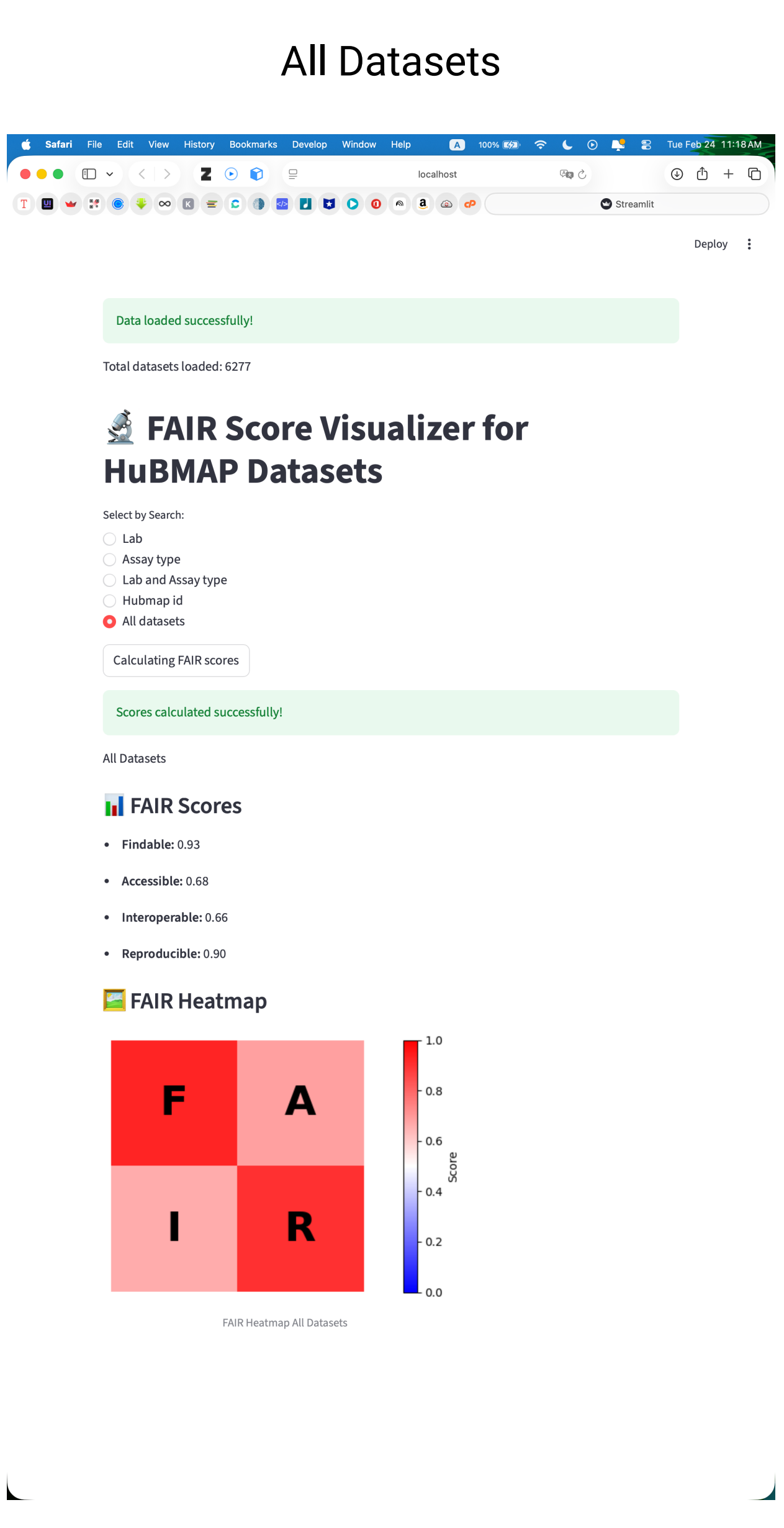


Conclusion

Implementation



Display FAIR Metrics



FUTURE WORK

- Adapting code to run automatically for the calculations (weekly or by weekly)
- Administration view/suggestion for weights
- Publicly accessible in git repo
- Implementing AI for a deep dive on the understanding Weight distribution based on the importance of the features
- Traing/Implement AI in a generic solution
- evaluation of systems
- metrics
- weights

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Python Libraries:

- typing
- random
- json
- os
- seaborn
- logging
- request
- Streamlit
- Hubmap_sdk
- pandas
- matplotlib
- numpy
- tqdm

Tools Used:

- github
- bridges-2
- python3